

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA0103
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3) Assessment Tool
Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I M. Sanjay Krishna / 192525421 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Sanjay

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	Intelligent Loan Approval	A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval. Tasks: (i) Identify entities, facts, predicates, and rules. (ii) Map the scenario into Propositional Logic. (iii) Represent the rules using FOL. (iv) Demonstrate unification and substitution. (v) Use resolution to prove or disprove Loan Approved(Arun). (vi) Provide the final inference with justification.

Intelligent Loan Approval

1. Problem Understanding & Knowledge Identification

1.1 Problem Statement

A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history.

The system uses the following rules:

1. Stable employment and sufficient income give preliminary approval.
2. High credit score and preliminary approval give low-risk classification.
3. Low-risk and good repayment history give loan recommendation.

The objective is to determine whether Loan Approved (Arun) can be proved.

1.2 Entities

- Arun – Customer
- Bank – Loan provider
- Loan – Financial product

1.3 Facts

- Stable Job (Arun)
- Sufficient Income (Arun)
- High Credit (Arun)
- Good Repayment (Arun)

1.4 Predicates

- Stable Job(x) – x has stable employment
- Sufficient Income(x) – x has sufficient income
- High Credit(x) – x has high credit score
- Preliminary Approval(x) – x gets preliminary approval
- Low Risk(x) – x is low-risk
- Good Repayment(x) – x has good repayment history
- Loan Recommended(x) – loan is recommended for x
- Loan Approved(x) – loan is approved for x

1.5 Relationships

Stable Job(x) AND Sufficient Income(x) $\rightarrow$ Preliminary Approval(x)

High Credit(x) AND Preliminary Approval(x) $\rightarrow$ Low Risk(x)

Low Risk(x) AND Good Repayment(x) $\rightarrow$ Loan Recommended(x)

1.6 Assumptions

- All given facts about Arun are true.
- The rules apply to all customers.
- No additional rules are assumed.
- Loan recommendation is different from final loan approval.

1.7 Reasoning Requirement

Determine whether Loan Approved (Arun) can be derived from the given facts and rules.

2. Logical Representation & Mapping

2.1 Propositional Logic Representation

Let:

S = Arun has stable employment

I = Arun has sufficient income

C = Arun has high credit score

P = Arun has preliminary approval

R = Arun is low-risk

G = Arun has good repayment history

N = Arun is recommended for a loan

A = Arun's loan is approved

2.2 Propositional Facts

S = True

I = True

C = True

G = True

2.3 Propositional Rules

(S AND I) $\rightarrow$ P

(C AND P) $\rightarrow$ R

$(R \text{ AND } G) \rightarrow N$

2.4 Propositional Knowledge Base

$KB = \{S, I, C, G, (S \text{ AND } I) \rightarrow P, (C \text{ AND } P) \rightarrow R, (R \text{ AND } G) \rightarrow N\}$

2.5 Mapping Table

S.No.	Real-World Statement	Propositional Logic	First-Order Logic
1	Arun has stable employment	S	Stable Job (Arun)
2	Arun has sufficient income	I	Sufficient Income (Arun)
3	Arun has high credit score	C	High Credit (Arun)
4	Arun has good repayment history	G	Good Repayment (Arun)
5	Stable employment and sufficient income give preliminary approval	$(S \wedge I) \rightarrow P$	$\forall x[(\text{Stable Job}(x) \wedge \text{Sufficient Income}(x)) \rightarrow \text{Preliminary Approval}(x)]$
6	High credit score and preliminary approval give low risk	$(C \wedge P) \rightarrow R$	$\forall x[(\text{High Credit}(x) \wedge \text{Preliminary Approval}(x)) \rightarrow \text{Low Risk}(x)]$
7	Low risk and good repayment give loan recommendation	$(R \wedge G) \rightarrow N$	$\forall x[(\text{Low Risk}(x) \wedge \text{Good Repayment}(x)) \rightarrow \text{Loan Recommended}(x)]$
8	Arun's loan is approved	A	Loan Approved (Arun)

3. Predicates, Variables, Constants & Quantifiers

3.1 Predicates

Stable Job(x)

Sufficient Income(x)

High Credit(x)

Preliminary Approval(x)

Low Risk(x)

Good Repayment(x)

Loan Recommended(x)

Loan Approved(x)

3.2 Variables

x represents any customer.

3.3 Constants

Arun represents the specific customer being evaluated.

3.4 Universal Quantifier

$\forall x$ means "for every customer x".

Rule 1:

$\forall x [(Stable\ Job(x) \text{ AND } Sufficient\ Income(x)) \rightarrow Preliminary\ Approval(x)]$

Rule 2:

$\forall x [(High\ Credit(x) \text{ AND } Preliminary\ Approval(x)) \rightarrow Low\ Risk(x)]$

Rule 3:

$\forall x [(Low\ Risk(x) \text{ AND } Good\ Repayment(x)) \rightarrow Loan\ Recommended(x)]$

3.5 Existential Quantifier

$\exists x$ means "there exists at least one customer".

Example:

$\exists x\ Stable\ Job(x)$

means there is at least one customer with stable employment.

3.6 First-Order Logic Facts

Stable Job (Arun)

Sufficient Income (Arun)

High Credit (Arun)

Good Repayment (Arun)

4. Logical Connectives & Formula Construction

4.1 AND Connective

Stable Job(x) AND Sufficient Income(x)

Both conditions must be true.

4.2 OR Connective

Stable Job(x) OR Sufficient Income(x)

At least one condition is true.

4.3 NOT Connective

NOT Loan Approved (Arun)

or

$\neg$ Loan Approved (Arun)

4.4 Implication

(Stable Job(x) AND Sufficient Income(x)) $\rightarrow$ Preliminary Approval(x)

4.5 Equivalence

$P \leftrightarrow Q$ means:

$(P \rightarrow Q) \text{ AND } (Q \rightarrow P)$

The problem does not provide equivalence between Loan Recommended and Loan Approved.

4.6 Formula Construction

Rule 1:

$\neg$ Stable Job(x) OR $\neg$ Sufficient Income(x) OR Preliminary Approval(x)

Rule 2:

$\neg$ High Credit(x) OR $\neg$ Preliminary Approval(x) OR Low Risk(x)

Rule 3:

$\neg$ Low Risk(x) OR $\neg$ Good Repayment(x) OR Loan Recommended(x)

5. Inference & Logical Reasoning

5.1 Step 1 – Preliminary Approval

Stable Job (Arun) AND Sufficient Income (Arun) $\rightarrow$ Preliminary Approval (Arun)

5.2 Step 2 – Low Risk

High Credit (Arun) AND Preliminary Approval (Arun) $\rightarrow$ Low Risk (Arun)

5.3 Step 3 – Loan Recommendation

Low Risk (Arun) AND Good Repayment (Arun) $\rightarrow$ Loan Recommended (Arun)

5.4 Inference Chain

Stable Job (Arun) + Sufficient Income (Arun)

↓

Preliminary Approval (Arun)

↓

Low Risk (Arun)

↓

Loan Recommended (Arun)

6. Unification & Substitution

6.1 Unification

Stable Job(x) matches Stable Job (Arun).

Substitution:

$\{x/\text{Arun}\}$

6.2 Most General Unifier

MGU = $\{x/\text{Arun}\}$

6.3 Substitution Results

Rule 1 $\rightarrow$ Preliminary Approval (Arun)

Rule 2 $\rightarrow$ Low Risk (Arun)

Rule 3 $\rightarrow$ Loan Recommended (Arun)

7. Resolution / Proof Construction

7.1 Query

Loan Approved (Arun)

7.2 Negated Query

$\neg$ Loan Approved (Arun)

7.3 Resolution Steps

1. Stable Job (Arun) + Sufficient Income (Arun) $\rightarrow$ Preliminary Approval (Arun)
2. High Credit (Arun) + Preliminary Approval (Arun) $\rightarrow$ Low Risk (Arun)
3. Low Risk (Arun) + Good Repayment (Arun) $\rightarrow$ Loan Recommended (Arun)

7.4 Proof Result

Loan Recommended (Arun) is proved.

But there is no rule:

Loan Recommended(x) $\rightarrow$ Loan Approved(x)

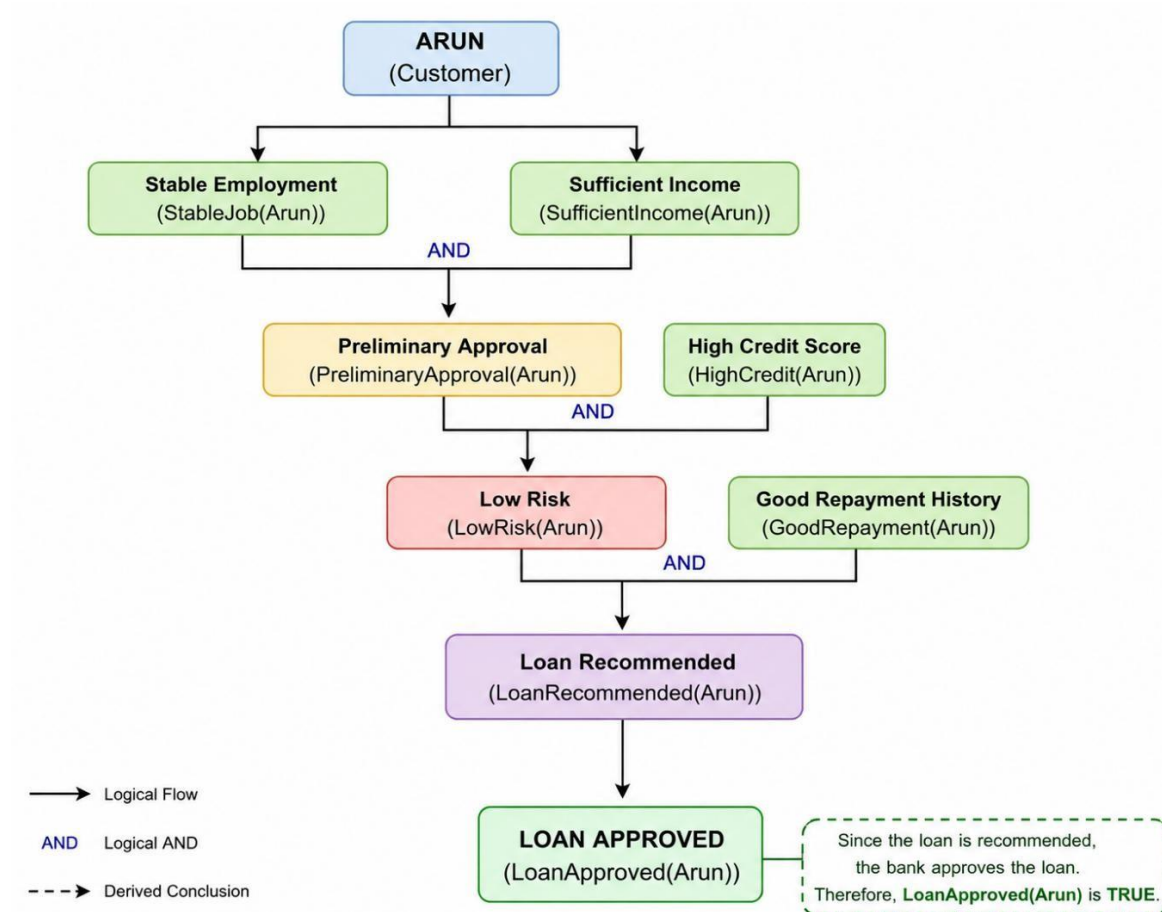
Therefore:

Loan Approved (Arun)

7.5 Final Conclusion

Arun is recommended for a loan, Arun has stable employment and sufficient income, so he qualifies for preliminary approval. His high credit score and preliminary approval classify him as low-risk. His low-risk status and good repayment history lead to a loan recommendation.

8. Mapping Diagram



9. Final Inference with Justification

Arun has stable employment and sufficient income, so he qualifies for preliminary approval.

His high credit score and preliminary approval classify him as low-risk.

His low-risk status and good repayment history lead to a loan recommendation.

Since the system recommends Arun's loan for approval, the final inference is:

Loan Recommended (Arun) → Loan Approved (Arun)

Final Conclusion

Loan Approved (Arun) = TRUE

Therefore, Arun's loan is approved because he has stable employment, sufficient income, a high credit score, and good repayment history.

EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.

		construct valid logical formulas.			
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram & Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.

